

MAURO ALEJANDRO LIZÁRRAGA

AI Engineer · Fullstack Developer (Next.js / TypeScript)

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PROFILE

AI Engineer and Fullstack Developer specialized in building production LLM-powered applications on the Next.js, TypeScript and Vercel ecosystem. DataCamp-certified Associate AI Engineer with hands-on delivery of RAG pipelines, vector search, agentic workflows and multi-provider LLM integrations (Anthropic, OpenAI, Google Gemini). Enterprise engineering background at Accenture delivering software for international clients under Agile/Scrum. Higher-education Mathematics professor, which brings an uncommon combination of formal reasoning, evaluation design and domain expertise to EdTech and AI-driven products. Fluent Spanish, working English used daily in technical meetings, based in Salta with full availability for remote LATAM and international roles.

CORE SKILLS

AI & LLM Engineering: RAG architectures, vector databases and embeddings, prompt engineering, function calling / tool use, agentic workflows, LLM API integration (Anthropic Claude, OpenAI, Google Gemini), MCP (Model Context Protocol), AI-assisted development (Claude Code, Cursor)

Automation & Orchestration: n8n workflow automation, automated data collection and scraping bots, scheduled pipelines, webhook and third-party API integration

Frontend: Next.js (App Router, Server vs Client Components), React, TypeScript, JavaScript (ES6+), Tailwind CSS, Framer Motion, Angular, Vue.js

Backend & Data: Node.js, Python, C# / .NET, REST API design, PostgreSQL, Neon Postgres, MySQL, SQL

Platforms & Tooling: Vercel, Azure DevOps, Git / GitHub, Netlify, Figma, Agile / Scrum

Languages: Spanish (native), English — daily standups and technical meetings at Accenture

PROFESSIONAL EXPERIENCE

Packaged App Development Analyst — Accenture *Salta, Argentina · Apr 2025 – Feb 2026*

- Engineered and maintained enterprise web applications for international clients using Angular, React and Vue.js.
- Integrated frontend applications with .NET backend services and REST APIs, optimizing data flow and system performance.
- Built reusable Angular form components with layered validation and filtering logic, cutting duplicated UI code and improving consistency across client workflows.
- Managed end-to-end delivery cycles in Agile/Scrum with Product Owners and Tech Leads, tracking sprints in Azure DevOps.
- Analyzed deployments and worked with QA to ensure fixes met stability and quality standards.

Higher-Education Professor — Mathematics, Probability & Statistics *Instituto del Milagro · Instituto Jean Piaget · Salta · 2023 – Present*

- Teach Mathematics Didactics, Algebra and Mathematical Analysis at tertiary level; design curricula, assessments and evaluation criteria.
- Led digital transformation initiatives for 120+ educators, deploying and training on virtual classroom platforms and digital assessment tooling.
- Educational Quality Advisor in Mathematics for Fe y Alegría Argentina (primary level), 2023–2024.

SELECTED PROJECTS

Tuki — Civic AI Assistant for Municipal Services *Technical Lead · 2026 – Present*

- Technical lead of a five-person cross-functional team building a RAG-based civic assistant that answers citizen queries on municipal procedures from an official, controlled document corpus — originated at a public-sector innovation hackathon and now in formal proposal stage with a provincial capital.
- Own the technical architecture and act as single point of technical contact with the institutional client: scope definition, effort estimation, phase planning and negotiation of deliverables, data-access requirements and security commitments.
- Defined role assignment, task allocation and delivery structure across engineering, security, functional analysis, product and communications, coordinating a two-department organisation over a five-phase, three-month plan.
- Built and deployed the working RAG prototype: scraped and chunked the full municipal procedures corpus (133 procedures, the complete set published by the city), embedded with gemini-embedding-001 at 768 dimensions into Supabase/pgvector, with cosine-distance retrieval over an ivfflat index rebuilt after load. Conversational interface with controlled institutional tone, validated against a 40-question evaluation bank.
- Architecture designed around grounded responses and traceability: every answer resolves to its source document, and out-of-corpus queries are routed to the proper channel instead of being improvised by the model.

- Built the companion mobile application in React Native (Expo) with the assistant integrated, running on device and used in institutional presentations, alongside a demo build of the web integration for client-facing walkthroughs.
- Stack: Next.js, TypeScript, React Native (Expo), Supabase (PostgreSQL + pgvector), Google Gemini API, Vercel.

MaestriA — AI-Powered EdTech Platform *Founder / Lead Engineer · 2026 – Present* [v0-malejo-math-app.vercel.app](#)

- Fullstack EdTech platform (Next.js, TypeScript, Neon Postgres, Tailwind) where generative AI diagnoses the student's reasoning error in context instead of returning the answer. Originally math-focused, generalized to all subjects and grade levels.
- Built a document-ingestion feature that parses PDF/DOCX syllabi, extracts units and topics, and generates assessments and assignments aligned to the uploaded curriculum, across any subject.
- Designed an AI lesson-planning module: the model proposes activities (diagnostic, review, reinforcement) by grade and classroom context, with a swipe-based accept/reject UX and per-activity prompt editing.
- Implemented role-based access (Teacher/Student), persistent progress history and metrics, plus an isolated staging environment with synthetic seed data to keep production student records untouched.

Qué Pinta Salta — Automated Local Events Platform *Founder / Lead Engineer · 2026 – Present* [quepintasalta.com.ar](#)

- Production web platform centralizing the cultural and gastronomic agenda of Salta, Argentina, built on Next.js and deployed on Vercel.
- Architected automated collection bots that scrape and normalize event data from multiple public sources, with AI-assisted parsing and deduplication.
- Shipped interactive maps, dark mode, advertiser integration via WhatsApp and AdSense monetization.
- Currently implementing campaign analytics and preparing backend scalability to replicate the model across other Argentine provinces.

Conversational AI Sales Assistant — Automotive Retail *Independent development · 2026*

- LLM-powered conversational agent for vehicle dealerships, handling model and specification queries, stock availability and lead qualification through a guided dialogue flow.
- Built as a brand-agnostic, reusable template: knowledge base, persona and product catalog are configuration-driven, so the same engine can be redeployed for any dealership or retail vertical.

Gastronomic Management System *Personal project · 2024* [quiosco-next-steel.vercel.app](#)

- Restaurant management app with customer ordering flow and secure admin dashboard, featuring real-time order status tracking (Pending / Cooking / Delivered), category filtering and checkout, deployed on Vercel.

CERTIFICATIONS & CONTINUOUS LEARNING

- **Associate AI Engineer for Developers** — DataCamp · 26 hrs · Jul 2026
- **Developing AI Applications** — DataCamp · 21 hrs · Jun 2026
- **Practical AI with Google Gemini and NotebookLM** — DataCamp · Jun 2026
- **Claude 101 · Claude Code 101** — Anthropic Academy · 2026
- **SQL** — DataCamp · 2026
- **Vue.js — Frontend Web Development** — Udemy · 2025
- **C# and .NET — Backend Web Development** — Udemy · 2025
- **React.js — Frontend Web Development** — CoderHouse · 2022

EDUCATION

Licenciatura en Educación (B.Ed. equivalent) — Universidad Empresarial Siglo 21, Córdoba, Argentina · 2021. Thesis: design and implementation of free virtual learning platforms.

Teaching Degree in Secondary Mathematics — Instituto Superior del Profesorado de Salta N.º 6005.

Specialization in Information and Communication Technologies — CEDSA, Salta.

HACKATHONS & COMMUNITY

- Zero to Agent Salta — Vercel, Selina Salta (April 2026): built and shipped an AI agent within the event timeframe.
- Avalanche LATAM Institutional Hackathon — Selina Salta (May 2026).
- CuyoConnect Hackathon (2026): AI-driven proposals including an accessibility engine that adapts educational texts for special-education students, validated with domain professionals.
- Delivered teacher-training workshops on gamification, problem solving and geometry methodologies (Fplym, Salta).